Sequences & Series

ANSWER KEY

Section 1: General Arithmetic & Geometric Progressions

1 In a geometric sequence, the 2nd term is 6 and the 5th term is 162. What is the common ratio r?

a. 2

b. 3 ✓

c. 4

d. 9

$\frac{a_5}{a_2} = \frac{ar^4}{ar} = r^3 = \frac{162}{6} = 27 \Rightarrow r = 3.$

2 Insert two geometric means between 5 and 320. What is the first inserted term?

a. 10

b. 20 ✓

c. 40

d. 80

Sequence has 4 terms: $a_1 = 5, a_4 = 320$. $r^3 = \frac{320}{5} = 64 \Rightarrow r = 4$. First mean $a_2 = 5 \times 4 = 20$.

3 Find the sum of all multiples of 4 between 10 and 100 inclusive.

a. $1,188$

b. $1,210$ ✓

c. $1,260$

d. $1,320$

First term $a = 12$, last term $l = 100$. Number of terms $n = \frac{100-12}{4} + 1 = 22 + 1 = 23$. Sum $S = \frac{23}{2}(12 + 100) = \frac{23}{2}(112) = 23 \times 56 = 1,288$.

Section 2: Sigma Notation & Special Sums

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4 Evaluate the sum: $\sum_{k=1}^{20} (3k + 2)$.

a. 650

b. 670 ✓

c. 690

d. 710

$\sum(3k + 2) = 3\sum_{k=1}^{20} k + \sum_{k=1}^{20} 2 = 3(\frac{20 \times 21}{2}) + 20(2) = 3(210) + 40 = 630 + 40 = 670.$

5 Using the formula $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$, evaluate $\sum_{k=1}^{10} k^2$.

a. 385 ✓

b. 420

c. 505

d. 550

$\frac{10(11)(21)}{6} = \frac{2310}{6} = 385.$

6 What is the sum of the telescoping series $\sum_{n=1}^{50} (\frac{1}{n} - \frac{1}{n+1})$?

a. $\frac{49}{50}$

b. $\frac{50}{51}$ ✓

c. 1

d. $\frac{51}{52}$

Terms cancel: $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \ \dots \ + (\frac{1}{50} - \frac{1}{51}) = 1 - \frac{1}{51} = \frac{50}{51}$.

Section 3: Infinite Series Convergence

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7 For what values of x does the geometric series $\sum_{n=0}^{\infty} (2x-3)^n$ converge?

a. $1 < x < 2$ ✓

b. $0 < x < 3$

c. $-1 < x < 1$

d. All real x

Convergence requires $|r| < 1$: $|2x-3| < 1 \Rightarrow -1 < 2x-3 < 1 \Rightarrow 2 < 2x < 4 \Rightarrow 1 < x < 2$.

8 What is the sum of the series $\sum_{n=1}^{\infty} \frac{3}{4^n}$?

a. $\frac{3}{4}$

b. 1 ✓

c. $\frac{4}{3}$

d. 2

First term $a = \frac{3}{4}$, common ratio $r = \frac{1}{4}$. Sum $= \frac{3/4}{1-1/4} = \frac{3/4}{3/4} = 1$.

9 Does the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ converge or diverge?

a. Converges to 1

b. Converges to $\pi^2/6$

c. **Diverges** ✓

d. Oscillates

The harmonic series is a standard p-series with $p = 1$, which diverges to infinity.

Section 4: Recursive Sequences & Induction

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10 A sequence is defined recursively by $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. What is the value of a_4?

a. 9

b. 15

c. 17 ✓

d. 21

$a_1 = 3$. $a_2 = 2(3) - 1 = 5$. $a_3 = 2(5) - 1 = 9$. $a_4 = 2(9) - 1 = 17$.

11 If $F_1 = 1, F_2 = 1,$ and $F_n = F_{n-1} + F_{n-2}$ represents the Fibonacci sequence, what is F_8?

a. 13

b. 21 ✓

c. 34

d. 55

Terms: $1, 1, 2, 3, 5, 8, 13, 21$. The 8th term is 21.

12 In a proof by mathematical induction, after verifying the base case $P(1)$, what is the inductive hypothesis?

a. **Assume $P(k)$ is true for some integer $k \geq 1$** ✓

b. Prove $P(1)$ is false

c. Assume $P(k+1)$ is true

d. Prove for all k simultaneously

The inductive hypothesis explicitly assumes the statement holds for an arbitrary integer $k \geq 1$ in order to prove $P(k+1)$.

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